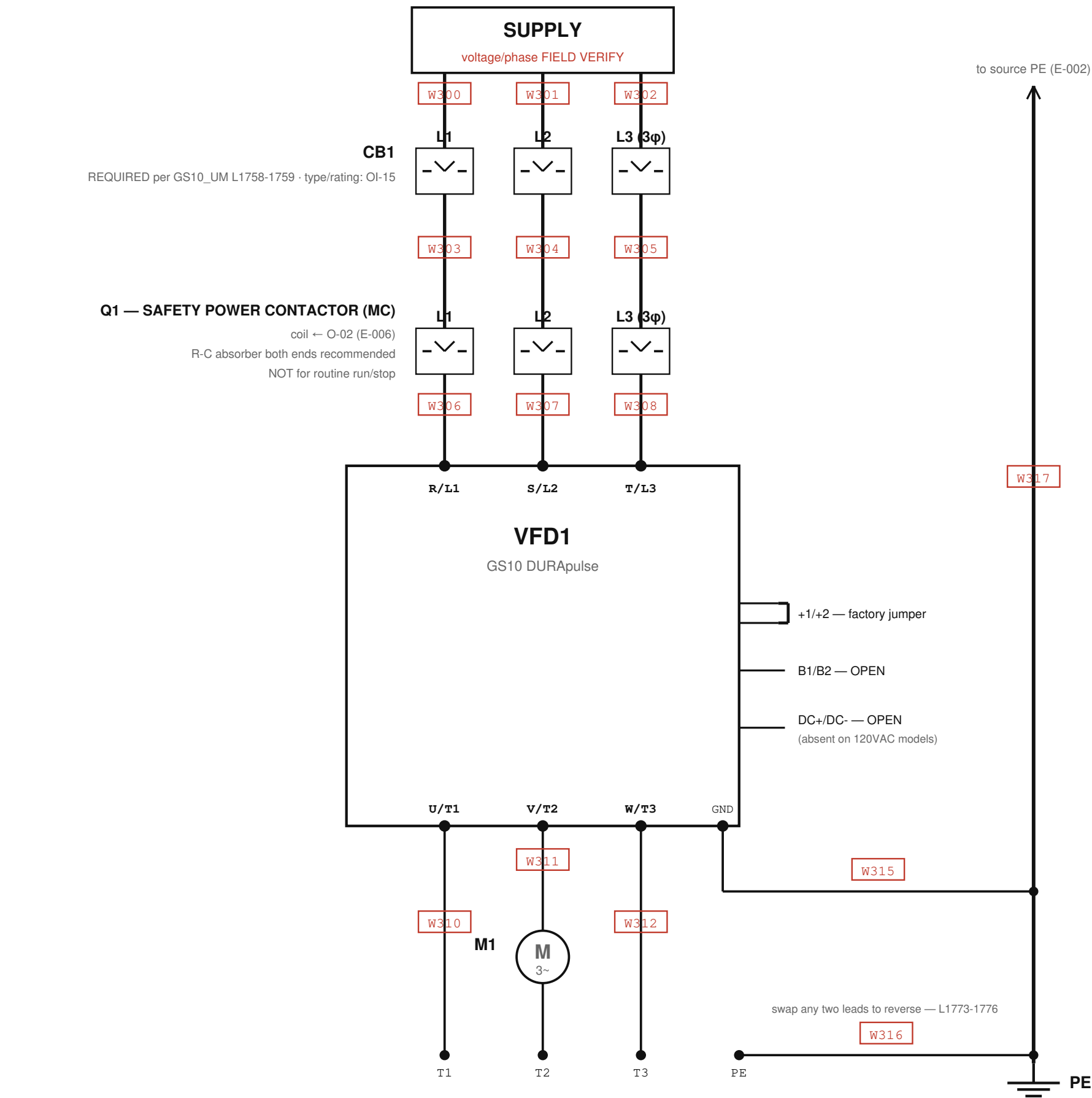


E-003 VFD POWER

Conv_Simple / CV-101 (garage conveyor) — supply → CB1 → Q1 (MC) → VFD1 (GS10) → M1 · every conductor FIELD VERIFY



CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W300	SUPPLY (E-002)	CB1.1	L1 supply	power_line	field_verify	
W301	SUPPLY (E-002)	CB1.3	L2/N supply	power_line	field_verify	
W302	SUPPLY (E-002)	CB1.5	L3 supply (if 3φ)	power_line	field_verify	phase count unknown
W303	CB1.2	Q1.1	L1 protected	power_line	field_verify	
W304	CB1.4	Q1.3	L2 protected	power_line	field_verify	
W305	CB1.6	Q1.5	L3 protected (if 3φ)	power_line	field_verify	phase count unknown
W306	Q1.2	VFD1.R/L1	L1 switched	power_line	field_verify	power ⊥ control — L1787
W307	Q1.4	VFD1.S/L2	L2 switched	power_line	field_verify	
W308	Q1.6	VFD1.T/L3	L3 switched (if 3φ)	power_line	field_verify	phase count unknown
W310	VFD1.U/T1	M1.T1	motor phase U	motor_lead	field_verify	
W311	VFD1.V/T2	M1.T2	motor phase V	motor_lead	field_verify	
W312	VFD1.W/T3	M1.T3	motor phase W	motor_lead	field_verify	
W315	VFD1.GND	PE bus	drive PE (≤0.1Ω)	ground	field_verify	
W316	M1.PE	PE bus	motor frame PE	ground	field_verify	
W317	PE bus	SUPPLY (E-002)	PE to source	ground	field_verify	

SAFETY NOTES

- Never start/stop via input power (L1811-1813); the MC (Q1) is emergency/safety switching only (L1754-1757).
- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1.
- PE resistance ≤0.1Ω (L1760-61); shield/conduit bonded both ends (L1792-95); RFI jumper in/out FIELD VERIFY (L1693-1718, OI-17).

———— VERIFIED — solid (no E-003 conductor qualifies yet; terminal names verified via table sources)
- - - - - FIELD VERIFY — dashed + red wire flag (all E-003 conductors; wire numbers PROPOSED — OI-19)

SOURCES: plc/GS10_UM.txt (terminals L1971-1986; topology L1750-1813) · plc/GS10_Integration_Guide.md (Q1 phase-5 test)
plc/Prog_init_ConvSimple_v2.1.st (vfd_run_permit) · plc/MIRA_PLC_WorkInstruction_v3.pdf (LOTO §2)

VFD POWER

Conv_Simple / CV-101 (garage conveyor)

MIRA / FactoryLM

terminals per GS10 UM 1st Ed Rev B;
MC placement per manual recommendation; as-built UNVERIFIED

SHEET

E-003

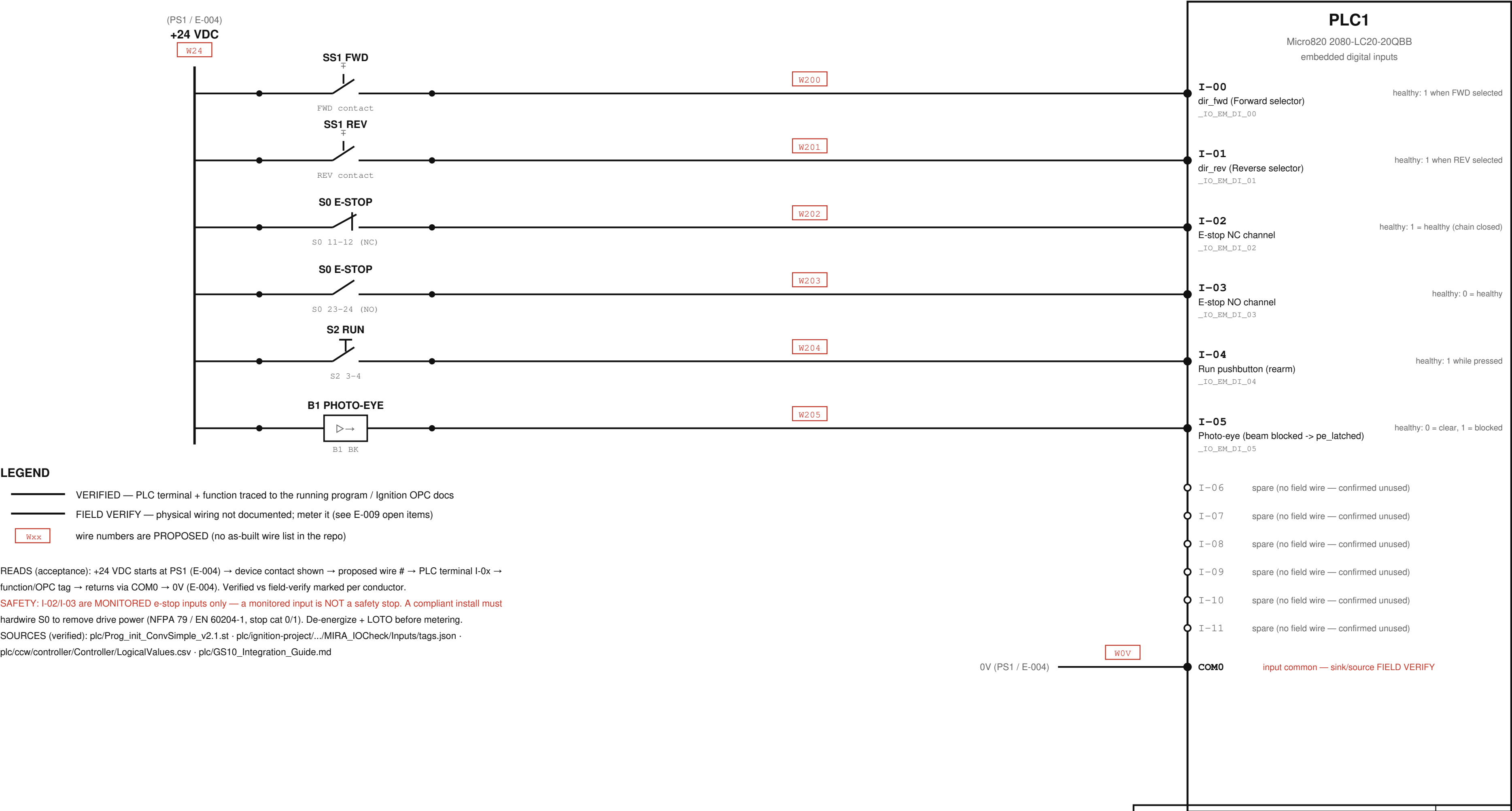
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3 of 9

E-005 PLC DIGITAL INPUTS

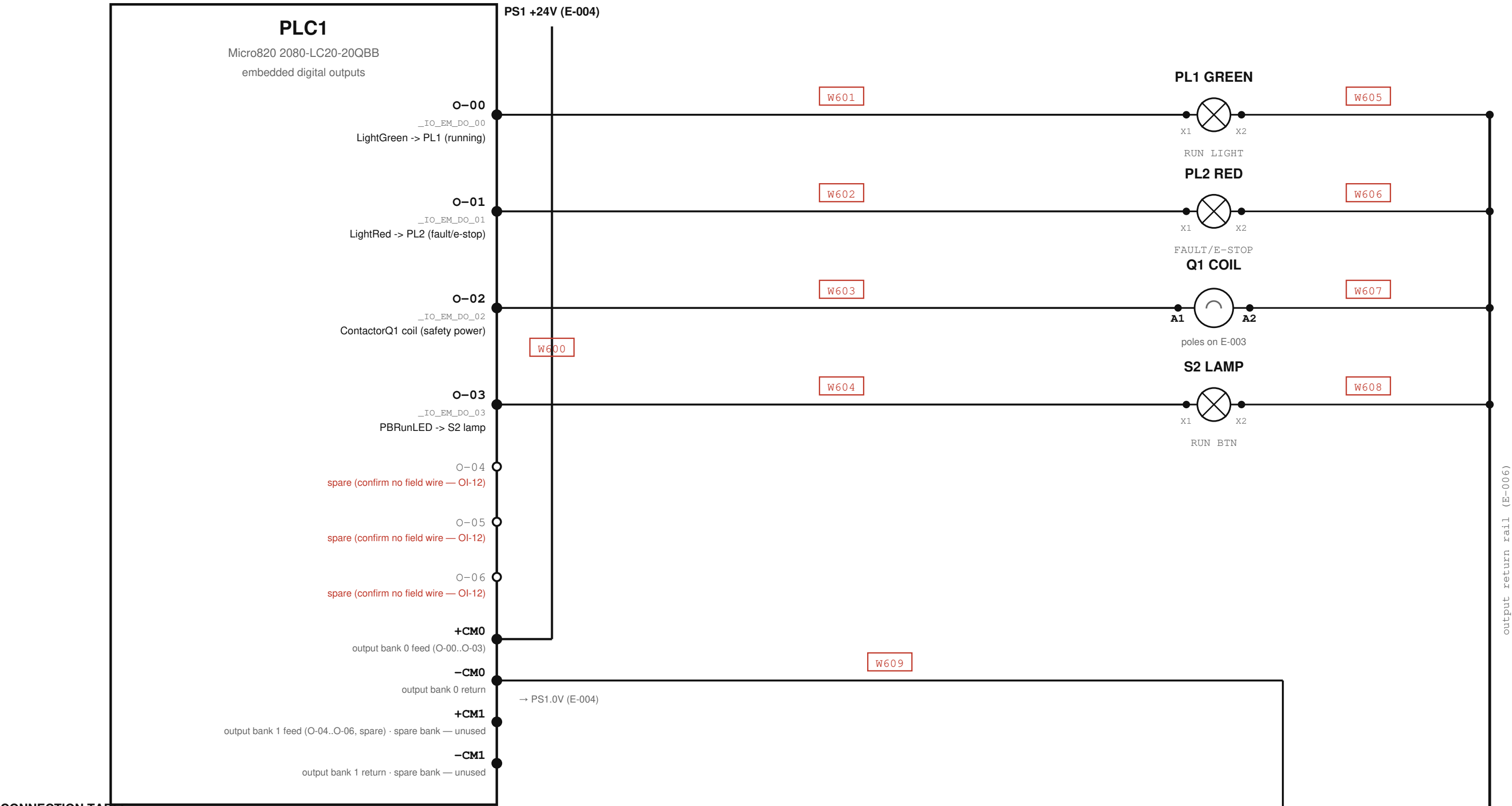
Conv_Simple / CV-101 (garage conveyor) — Micro820 2080-LC20-20QBB embedded DI (24 VDC)



PLC DIGITAL INPUTS Conv_Simple / CV-101 (garage conveyor)		SHEET E-005
MIRA / FactoryLM Drawn: MIRA / FactoryLM		REV A 2026-07-10 5 of 9

E-006 PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor) — O-00..O-06 · pilots, contactor coil, button lamp · GS10 control = Modbus (E-007), NO GS10 DI wiring



CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W600	PS1.+24V	PLC1.+CM0	output bank 0 feed	power_24vdc	field_verify	
W601	PLC1.O-00	PL1.X1	LightGreen (run light)	control_output	field_verify	
W602	PLC1.O-01	PL2.X1	LightRed (fault/e-stop light)	control_output	field_verify	
W603	PLC1.O-02	Q1.A1	ContactorQ1 coil (safety power)	control_output	field_verify	
W604	PLC1.O-03	S2.X1	PBRunLED (run button lamp)	control_output	field_verify	
W605	PL1.X2	output return rail (E-006)	run light return	control_return	field_verify	
W606	PL2.X2	output return rail (E-006)	fault light return	control_return	field_verify	
W607	Q1.A2	output return rail (E-006)	contactor coil return	control_return	field_verify	
W608	S2.X2	output return rail (E-006)	PB lamp return	control_return	field_verify	
W609	PLC1.-CM0	PS1.0V	output bank 0 return	power_0v	field_verify	

Hardwired fallback documented in MIRA-WI-001 v3 §3
Fig 1 (DO_03..DO_07 → GS10 DI1..DI5 + DCM) — NOT ACTIVE and NOT drawn: P02.0x at factory default per the 2026-05-20 parameter export; DO_07 does not exist on the 2080-LC20-20QBB (O-00..O-06); DO_03 collides with PBRunLED in the live I/O map. Bench presence of any such wiring = FIELD VERIFY (OI-18).

MODBUS CROSS-REF (run/dir/freq):

Run/dir/freq commands reach VFD1 over RS-485 (E-007): 0x2000 cmd (STOP=1, FWD+RUN=18, REV+RUN=34), 0x2001 freq. P00.20=1, P00.21=2 (RS-485) verified by 2026-05-20 parameter export.

SAFETY NOTE:

Monitored outputs are NOT a safety function. E-stop must remove power per NFPA 79.

Q1 drop path is the power-removal; verify it works.

FIELD VERIFY — dashed (no as-built; O-00/O-01/O-03 from CCW v4.0; O-02 corroborated)

SOURCES:

- CCW_VARIABLES_v4.0.txt
- Prog_init_ConvSimple_v2.1.st
- MIRA_PLC_WorkInstruction_v3.pdf
- GS10_actual_parameters_5.20.26.xlsx
- GS10_UM.txt

PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor)

MIRA / FactoryLM

output map CCW v4.0 + live Prog_init; O-02 do-not-reuse

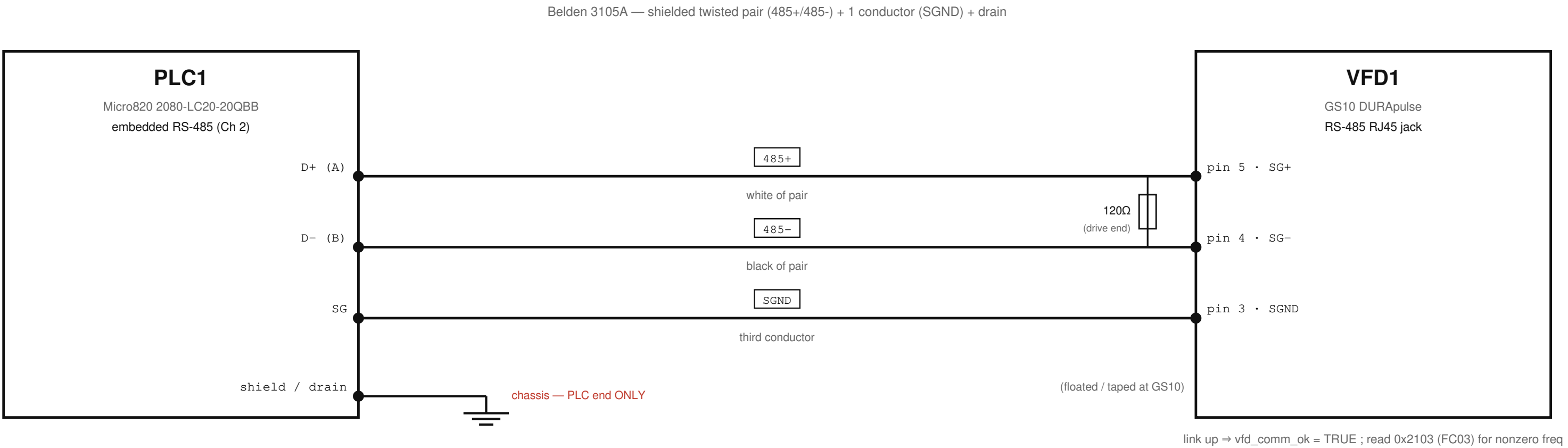
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REV A
2026-07-10

6 of 9

E-007 RS-485 / MODBUS RTU COMMUNICATION

Conv_Simple / CV-101 (garage conveyor) — Micro820 (master) ↔ GS10 (node 1) · Modbus RTU only



CONNECTION TABLE

Src dev	Src terminal	Dst dev	Dst terminal/pin	Cable / conductor	Wire	Evidence	Notes
PLC1	D+ (A)	VFD1	RJ45 pin 5 / SG+	Belden 3105A pair · white of pair	485+	verified	Polarity: if silent/garbled, swap 485+/485- first.
PLC1	D- (B)	VFD1	RJ45 pin 4 / SG-	Belden 3105A pair · black of pair	485-	verified	
PLC1	SG	VFD1	RJ45 pin 3 / SGND	3rd conductor · third conductor	SGND	verified	Dedicated wire, NOT the shield.
PLC1	shield / chassis	VFD1	(floated)	tinned drain · drain	SH	field_verify	Land drain at PLC end ONLY; tape/float at GS10 end.

CCW SERIAL PORT: Modbus RTU, Master · 9600 · 8N1 (P09.04 = 12) · Channel 2 · Node 1 (P09.00 = 1)

Older GS10_Integration_Guide.md said 8N2 (P09.04=13) - SUPERSEDED by bench-verified 8N1.

CORRECTED from May-16 draft: Channel 0 → 2 · SGND pin 1/8 → pin 3 · 8N2 → 8N1. Do not copy the old values.

TROUBLESHOOTING (this circuit)

- Nothing after config -> swap 485+/485- (A/B polarity; vendors disagree).
- CRC errors -> usually polarity. Complete silence -> usually baud mismatch (confirm 9600 8N1).
- GS10 CE10 / F30 comm-timeout -> no Modbus packets; check wiring, baud, P09.03 timeout.
- 120 ohm termination across SG+/SG- at the GS10 end for long runs (bench <2 m usually fine).
- SGND is a dedicated conductor, not the shield; missing common -> intermittent corruption.

- VERIFIED (recovered CommsToVFD + Beginner_Verify p48 + Prog_init v2.1)
- FIELD VERIFY (exact PLC chassis ground point; cable P/N if not Belden 3105A)

SOURCES: CCW/MIRA_PLC/docs/instructions/Conv_Simple_CommsToVFD.pdf §2 (recovered style + wiring)
CCW/MIRA_PLC/docs/instructions/Conv_Simple_GS10_Beginner_Verify_V2.pdf p48 (corrected pins/channel)
plc/Prog_init_ConvSimple_v2.1.st (Channel 2, node 1, 0x2103)
plc/GS10_Integration_Guide.md §3 (RJ45 pinout, register map)

RS-485 / MODBUS RTU Conv_Simple / CV-101 (garage conveyor)	SHEET E-007
MIRA / FactoryLM recovers MIRA-WI-001 / Conv_Simple_CommsToVFD §2	REV A 2026-07-10 7 of 9